RAM PRASATH K

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LinkedIn | GitHub | Notion Portfolio

PROFESSIONAL SUMMARY

ML Engineer with 2+ years of experience specializing in production LLM/VLM inference, multi-agent systems, and edge AI deployment. Shipped vLLM-served Vision-Language pipelines with W4A16/W8A8 quantization, architected multi-agent platforms on Google ADK + MCP implementing ReAct, self-reflection, and hierarchical delegation patterns, and deployed optimized speech models on NXP edge hardware at Bosch. Currently building diffusion-transformer Vision-Action models at Ubisoft India Studios. Deep hands-on experience across SFT, LoRA, QLoRA, DPO, and GRPO fine-tuning on RTX 4090, RTX Pro 6000, and AMD MI300X. **Won #1 at the Mistral AI Worldwide Hackathon 2026**. Targeting inference engineering, agentic AI, and MLOps roles.

EXPERIENCE

Jr. R&D Engineer — Ubisoft India Studios — Oct 2025 – Present

Game AI & Quality Control division — building production Vision-Action models for autonomous gameplay agents

- Architected a diffusion-transformer Vision-Action framework for long-horizon decision-making, enabling autonomous gameplay agents with stable game control — applicable to multimodal model training and evaluation pipelines.
- Partnered directly with gameplay and QA engineering teams to scope requirements, deliver internal tooling, and iterate on vision-driven automation systems based on production feedback.

Stack: PyTorch · Diffusion Transformers · C# · C++ · Multimodal Vision-Action Models

Data Science Engineer — Ninestars Information Technologies — Jul 2024 – Sep 2025

R&D division focused on production AI systems for OCR, document intelligence, and multi-agent automation

AI-Driven OCR & Document Understanding

- Optimized Vision-Language Model inference using vLLM with W8A8 / W4A16 quantization on RTX 4090; profiled tokens/sec, TTFT, cost-per-request, and throughput trade-offs to deliver stable low-latency production serving under tight GPU memory constraints.
- Developed end-to-end OCR and document understanding pipelines using Qwen-VL / Qwen2.5-VL and a custom Seq2Seq Transformer for OCR correction, NER, and structured extraction — cutting manual review by 20% and boosting entity extraction precision by 15%.
- Built a layout-aware annotation acceleration system using YOLO and RT-DETR integrated with a Flask-based backend, achieving a 4× increase in labeling throughput; partnered with the annotation team to scope requirements and iterate on delivery.

Multi-Agent Platform

- Architected a multi-agent automation platform on Google ADK + MCP, integrating locally-served GPT-OSS-20B, Ollama, WebSockets, REST APIs, and AWS-hosted vLLM — implementing ReAct reasoning, self-reflection loops, and hierarchical agent delegation patterns across the orchestration layer.
- Built evaluation pipelines and observability hooks for agentic workflows, tracking accuracy, latency, and tool-call success rates against production readiness thresholds using Weights & Biases.

Stack: Python · PyTorch · Hugging Face · vLLM · Google ADK · MCP · GPT-OSS-20B · Ollama · AWS · FastAPI · Flask · WebSockets · YOLO · RT-DETR · PostgreSQL · W&B

AI/ML Research Engineer (Intern) — Bosch Global Software Technologies — Aug 2023 – Jul 2024

HMI division — 11-month end-to-end ownership from research to NXP edge deployment

Voice Biometrics

- Built a ResNet-SE automotive voice authentication system with anti-spoofing — achieved 95% accuracy and 4% Equal Error Rate (EER) under real-world noise conditions.
- Trained and deployed multilingual speech models (VoxCeleb, IndicSuperb datasets); exported optimized inference via ONNX / TensorFlow Lite for NXP edge hardware, owning the complete research-to-edge-deployment lifecycle.
- Collaborated with HMI and embedded engineering teams to scope latency and memory budgets, and shipped validated models onto target hardware.

Stack: PyTorch · Torchaudio · ResNet-SE · ONNX · TensorFlow Lite · NXP Edge Hardware

TECHNICAL SKILLS

Languages: Python, C++, C#, SQL

ML Engineering & Serving: vLLM, FastAPI, Flask, REST APIs, Docker, AWS, MCP Servers, OpenAI-compatible APIs

Agentic Systems: Google ADK, MCP, ReAct, Self-Reflection, Hierarchical Delegation, Multi-Agent Orchestration, Tool-Use, RAG

Model Optimization: W8A8, W4A16, AWQ, ONNX, TensorFlow Lite, Knowledge Distillation, Quantization-Aware Training

Training & Fine-tuning: PyTorch, Hugging Face Transformers, SFT, LoRA, QLoRA, DPO, GRPO, FSDP, Distributed Training

Evaluation & Observability: LLM-native metrics (tokens/sec, TTFT, cost-per-request), Eval Pipelines, W&B Tracing, LLM-as-judge

Data & Storage: PostgreSQL, pgvector, GraphDB, Neo4j, RAG, Vector Stores (Qdrant)

MLOps & Tooling: Weights & Biases, Git, CI/CD, Linux, Docker

ACHIEVEMENTS

- Mistral AI Worldwide Hackathon 2026 — Winner, #1 (Online Track): Built Mistral Raid, an AI-powered dungeon crawler where the boss dynamically adapts to player voice and gameplay using Mistral LLM agents, Voxtral STT, and real-time combat telemetry.
- AMD × Unsloth × PyTorch Synthetic Data AI Agents Challenge (Nov 2025): Designed and fine-tuned QA AI agents using synthetic data pipelines and Synthetic-Data-Kit on AMD Instinct MI300X GPUs, exploring SFT, LoRA, QLoRA, and DPO fine-tuning workflows on next-gen GPU architecture.
- NVIDIA Nemotron Model Reasoning Challenge — Kaggle (2025–2026): Improved Nemotron 3 Nano reasoning accuracy to 85%, using prompting, synthetic data generation, data curation, and fine-tuning (SFT, LoRA, GRPO) on RTX Pro 6000.

PROJECTS

XFINITE-OCR Professional — Production Document Intelligence System

- Designed and deployed a vLLM-powered Vision-Language Model (VLM) inference system with an OpenAI-compatible API, enabling scalable multimodal document understanding under tight GPU memory constraints.
- Implemented decoding-time inference controls to improve OCR robustness; conducted inference profiling studies tracking tokens/sec, TTFT, memory usage, and throughput trade-offs across batch sizes and context lengths.

Multimodal Video Captioning System with Commonsense Reasoning

- Developed a multimodal captioning system integrating visual representations and textual priors via a hybrid ResNet152 + Transformer decoder architecture, enhancing semantic relevance and caption coherence.

Student Evaluation Framework with RAG & GraphDB

- Designed a memory-augmented evaluation system combining Retrieval-Augmented Generation (RAG) with a Graph Database, enabling structured knowledge retrieval and relational reasoning — improving response consistency over flat vector-only retrieval.

PUBLICATIONS & PATENT

- Appearance-Trajectory Network for Video Anomaly Detection System — ICCCNT 2024, IEEE, IIT Mandi (July 2024): Memory-augmented CNN for unsupervised video surveillance achieving 82% AUC (appearance) and 84.9% AUC (skeleton), with integrated video captioning for explainability.
- Simultaneous Voice Authentication and Decision-Making Based on Intent & Entity Classification — Patent No. 202441102999 (filed Dec 24, 2024): System combining voice authentication with real-time intent and entity classification for secure, context-aware voice-driven decision automation.

EDUCATION

Indian Institute of Technology – Madras (IIT-Madras) Dec 2024
Bachelor of Science in Data Science and Applications (Online / Distance programme)

Amrita Vishwa Vidyapeetham, School of Artificial Intelligence Jun 2024
Master of Technology in Data Science — CGPA 8.5 / 10

SASTRA Deemed University Jun 2022
Bachelor of Technology in Mechanical Engineering